THINKIN’ ROCKS · NERD OUT III

What if your wearables shared a clock?

The health insights between the sensors

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SensWear · 20 October 2026



Concept illustration

Three devices.
One body.

Finger	Pulse waveform
Wrist	Pulse + movement
Ear	Movement + possible cardiorespiratory sensing

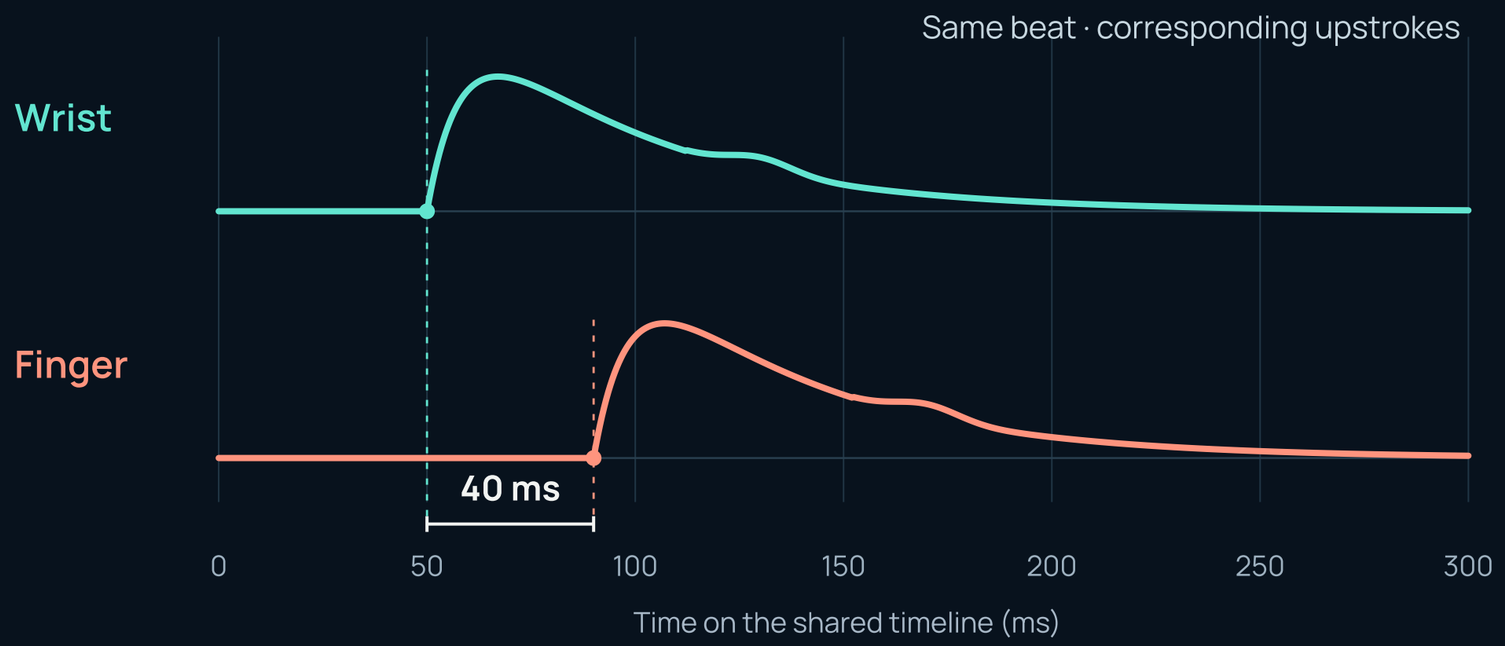
Their relationship needs **shared time.**



Future network concept. Sensor capabilities and raw-data access vary by device. Ear physiology here refers to research hearables.

PULSE TRANSIT TIME

One heartbeat.
Two arrival times.



Pulse-wave velocity

$$\frac{\text{arterial path length}}{\text{transit time}}$$

0.20 m / 0.040 s

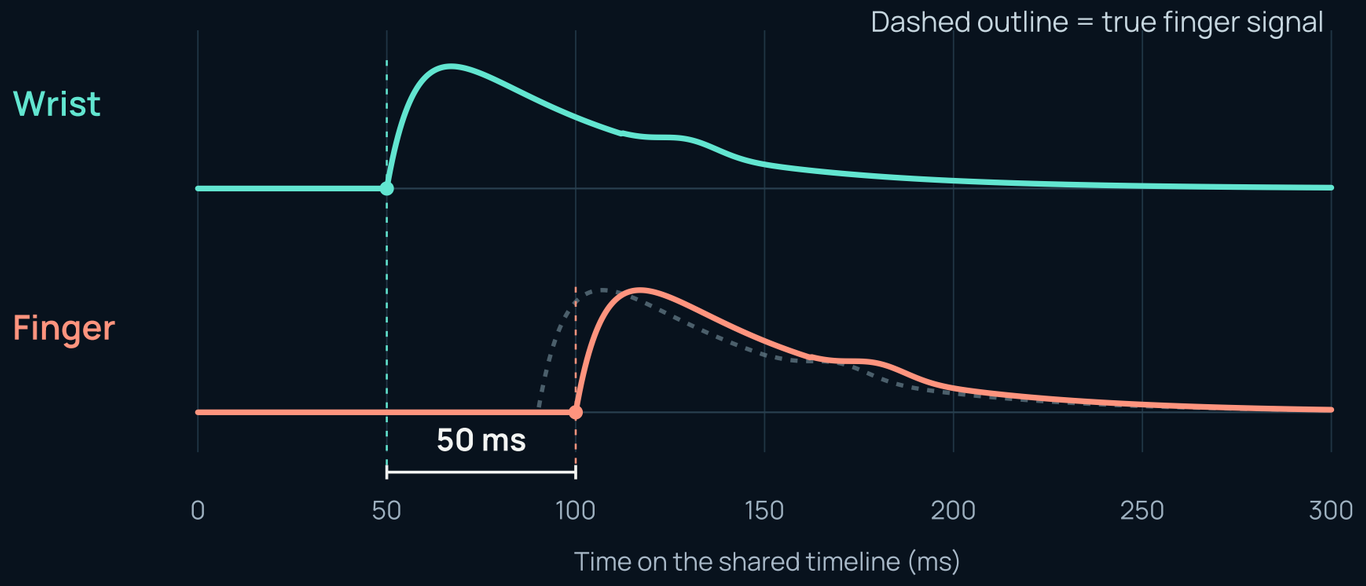
5.0 m/s

A pressure wave travels along the artery.

Illustrative signals and assumed path. Compare corresponding pulse upstrokes. This is wave speed, not blood-cell speed.

TRY IT · CLOCK ERROR

10 milliseconds can change the answer



Finger clock offset

+10 ms

Apparent transit time

50 ms

Estimated PWV

4.0 m/s

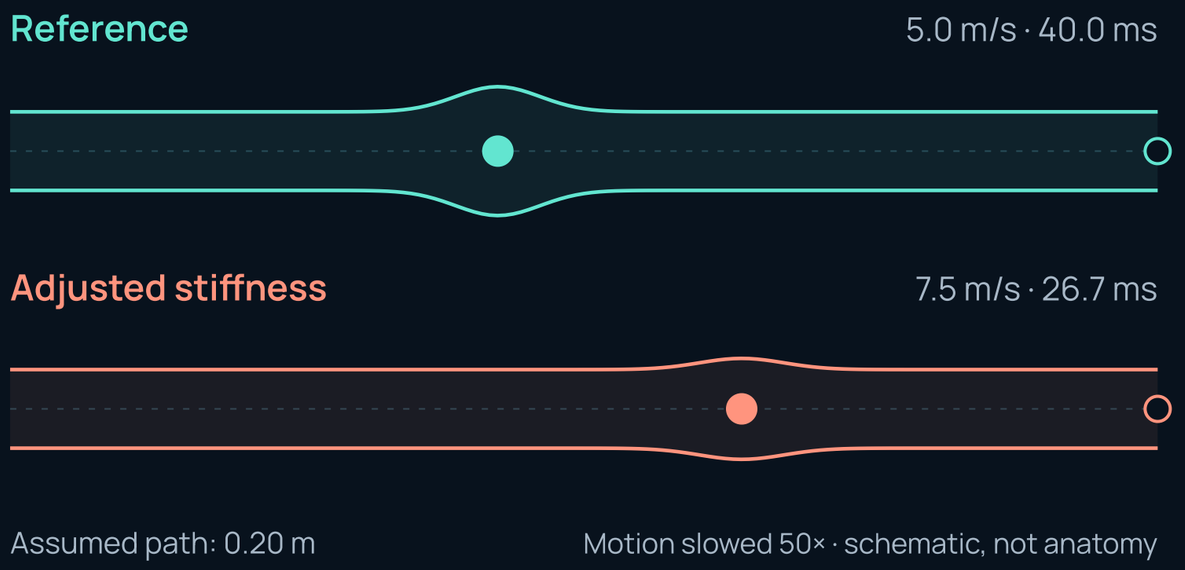
20% below the true value

True scenario: 40 ms · 5.0 m/s

Synthetic example. The body stays the same. Only the clock changes. Assumed arterial path: 0.20 m.

ARTERIAL STIFFNESS

Stiffer arteries carry a faster wave



Same path · earlier arrival

7.5 m/s

26.7 ms transit

PWV can reveal **stiffness-related changes.**

Relative wall stiffness **2.25×**

Idealized tube model: PWV ∝ √stiffness, with pressure and geometry fixed. Real arteries also depend on pressure, tone and geometry.

RESEARCH POSSIBILITIES

New insights live in the relationships

01 Vascular dynamics

How does pulse transit change within a person over time?

Pulse at wrist + pulse at finger

02 Regional circulation

Do timing and waveforms differ across locations or sides?

Multiple optical pulse sites

03 Measurement confidence

Is that change physiological, or is the wrist moving?

Pulse + motion + another pulse site

04 Recovery in context

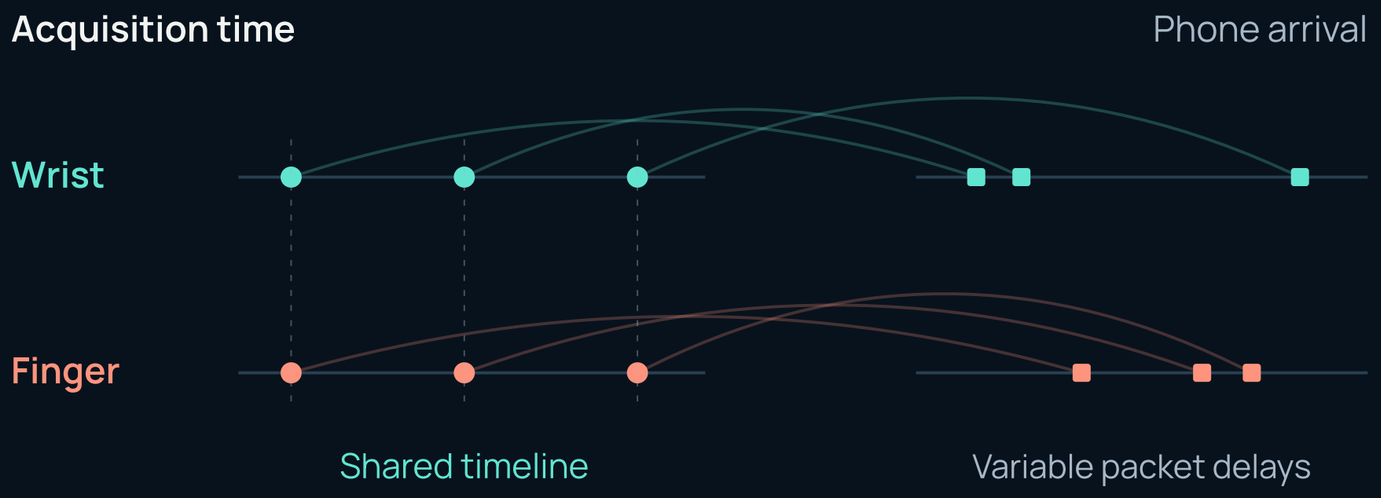
How do pulse dynamics relate to breathing and activity?

Cardiac timing + respiration + movement

Research questions, not validated diagnoses. Each use case needs suitable sensors, timing precision and independent validation.

THE ENGINEERING CHALLENGE

A shared clock starts at the sensor



Schematic transport example; independent of physiological pulse delay

estimated shared time = rate × local time + offset

01 Timestamp acquisition

Phone arrival includes radio and buffer delays.

02 Correct offset and drift

Clocks keep drifting after the first sync.

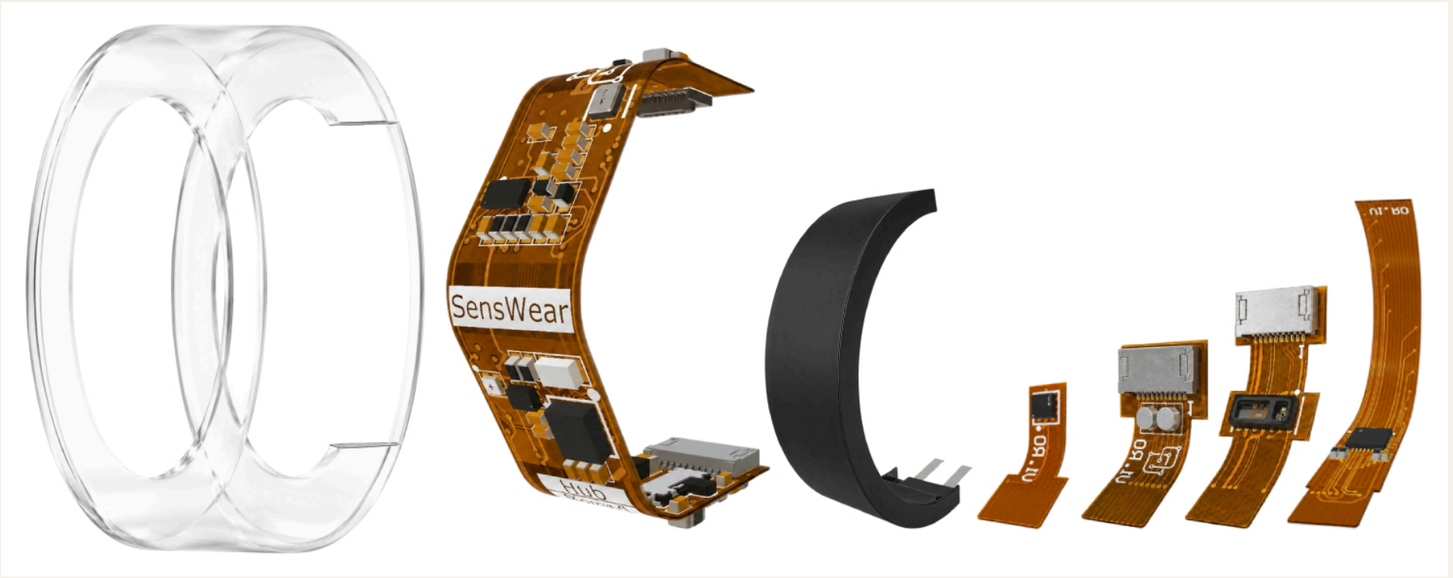
03 Measure the whole error

Sampling, filters, placement and signal quality.

100 Hz gives 10 ms between samples. Interpolation can refine timing, but cannot recover information the measurement lost.

SensWear

A platform for the experiment



Modular sensors

PPG, motion and temperature

Raw-data access

Inspect signals and timestamps

Open hardware + firmware

Develop the timing protocol

> **The next experiment**
>
> Build and validate precise synchronization between devices.

A RESEARCH INVITATION

The first experiment: two sites, one pulse

01 **Measure the clocks**

Apply a common reference event.
Quantify residual error and drift.

02 **Record wrist + finger PPG**

Verify sample times, filtering, placement and arterial path.

03 **Compare with a reference**

Repeat under controlled conditions.
Report uncertainty and failed readings.

The question

Can we see a physiological change larger than our timing error?

PWV can support stiffness research. Blood-pressure estimation needs its own calibration and validation.

SENSWEAR

What would you measure together?

Let’s build the experiment.



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Sources & technical notes ↗